FEAT-ZD-48600 · DS-04 Identify the components/layers to be changed · Components, layers, sources or pipelines to change

Data Science agent · item task.components · source hobasa-data-science-agent/outputs/FEAT-ZD-48600/task.components.md · approved by automated checks and independent judge

task.components: FEAT-ZD-48600

Components and layers the Data Science lane creates or updates for FEAT-ZD-48600 ("Show which overdue invoices belong to vendors that usually pay late"). This is DS-04 Identify the components/layers to be changed. The workbook outcome is "the list of components/layers needs to be created/updated with minimum lines of code". This file is a plan for DS-05 (coding) and DS-06 (unit tests). It changes no code.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	DS-04 Identify the components/layers to be changed (attempt 1)	record:brief
Owner / gate	Developer / no gate: automated checks, the independent judge and sampled human evaluation	record:brief
Inputs	feat.tasks (b0fd4d2), feat.stories (0590ea4), feat.prd (908b58d), feat.trd (3637e51), feat.design (722a98f), task.repo_access (daae080)	record:brief
Background read	task.secrets_scope (DS-02), task.db_changes (DS-03), feat.clarifications (DES-03)	journey:task.secrets_scope; journey:task.db_changes; journey:feat.clarifications
Lane tasks	TSK-03 (STORY-03, `risk.vendor_risk`) and TSK-04 (STORY-04, `explain.build_explanation_request(..., risk=None)`)	feat.tasks (sections 1, 3)
Repository	`ap_forecast`, Python 3.9 standard library, branch `feature/story`, CI `/usr/bin/python3 ci.py`	task.repo_access (section 1); file:ap_forecast/CLAUDE.md
Code read	`src/ap_forecast/__init__.py`, `predict.py`, `explain.py`, `models.json`; `tests/test_forecast.py`; `ci.py`; `README.md`	tool:repo; file:ap_forecast/ci.py; file:ap_forecast/README.md

1. Result

Five files change in one repository: one new source module, two existing source modules, one new test file and one existing test file. No other layer changes: no data store, no configuration, no CI script, no secret, no other repository.

#	Component	Layer	Kind	Task	Source
C-1	`src/ap_forecast/risk.py` with `HIGH_ABOVE_DAYS`, `MEDIUM_ABOVE_DAYS`, `vendor_risk`	Rule (domain logic, no LLM)	New module	TSK-03	feat.design (HC-4, B2.1, B2.2, DD-9); feat.tasks (TSK-03)

#	Component	Layer	Kind	Task	Source
C-2	`src/ap_forecast/explain.py`: `build_explanation_request` gains `risk=None` as the last parameter	LLM request builder	Changed, additive	TSK-04	feat.design (HC-5, B2.1, B2.2); feat.trd (C-5, IF-4.1)
C-3	`src/ap_forecast/__init__.py`: import `vendor_risk`, add it to `__all__`, version step from `0.9.0`	Package public API	Changed, additive	TSK-03 (owns the version step)	feat.design (B2.1, DD-11); feat.tasks (section 2, TSK-03)
C-4	`tests/test_risk.py`	Unit tests	New file	TSK-03	feat.tasks (section 2, TSK-03); feat.design (B2.5)
C-5	`tests/test_forecast.py`: two new tests; existing tests unchanged	Unit tests	Changed, additive	TSK-04	feat.tasks (section 2, TSK-04); feat.design (B2.5)

```
flowchart LR
  subgraph PKG["ap_forecast (Python 3.9)"]
    P["predict.predict_delay_days<br/>(unchanged)"] --> R["risk.vendor_risk<br/>C-1 new"]
    E["explain.build_explanation_request<br/>+ risk=None (C-2)"]
    M["models.json<br/>(unchanged)"] --> E
    I["__init__.py<br/>+ vendor_risk, version (C-3)"]
    R --> I
    P --> I
    E --> I
  end
  T1["tests/test_risk.py (C-4)"] -.-> R
  T2["tests/test_forecast.py (C-5)"] -.-> E
```

Sources: feat.design (A2 diagram, B2.1) for the new and changed components; tool:repo for the existing imports (`models.json` → `explain`, modules → `__init__`). The test edges are this plan's mapping from feat.tasks (section 2).

2. How the existing code works (read before changing it)

[['Module', 'What it does today', 'What this feature relies on', 'Source'], ['`predict.py`', '`predict_delay_days(history)` returns the rou

Blast radius. Inside `ap_forecast`, the only callers of `build_explanation_request` and `predict_delay_days` are the `__init__.py` re-export and `tests/test_forecast.py` (mcp:codegraph). No other lane imports the `ap_forecast` package; the other lanes build against the DC-2 and DC-3 contracts (journey:feat.clarifications CL-X2; feat.design A3). Nothing new calls `build_explanation_request` with `risk`; the end-to-end wiring uses only `vendor_risk` (journey:feat.clarifications CL-DS6; feat.design B5).

3. Change per component, with the minimum code

The sketches below are the smallest change that meets the design. They are a plan for DS-05, which writes the final code and comments.

C-1 `src/ap_forecast/risk.py` *(new, TSK-03)*

```
"""Rule-based vendor payment-risk level from the predicted delay. No LLM, no network, no config read."""
from typing import Iterable

from .predict import predict_delay_days

HIGH_ABOVE_DAYS = 30
MEDIUM_ABOVE_DAYS = 10
```

```
def vendor_risk(history: Iterable[int]) -> str:
    """'high' if the predicted delay is above HIGH_ABOVE_DAYS, 'medium' if above MEDIUM_ABOVE_DAYS, else 'low'."""
    days = predict_delay_days(history)
    if days > HIGH_ABOVE_DAYS:
        return "high"
    if days > MEDIUM_ABOVE_DAYS:
        return "medium"
    return "low"
```

Rule	Why	Source
Call `predict_delay_days` exactly once and keep the result	A generator `history` is consumed on the first pass	journey:feat.clarifications (CL-DS2); feat.tasks (TSK-03)
Strict > on the rounded prediction; constants 30 and 10	AC-03.1 to AC-03.4: exactly 30 → `'medium'`, exactly 10 → `'low'`	feat.design (DD-9, B2.2); feat.stories (AC-03.1 to AC-03.4); feat.trd (IF-3.1)
Import only `predict_delay_days`, never `explain`	The rule cannot reach the LLM path (AC-03.7)	feat.design (B2.2); journey:feat.clarifications (CL-DS4); feat.stories (AC-03.7)
No validation of history values	`predict_delay_days` already applies `int(d)`	journey:feat.clarifications (CL-DS3); feat.design (B2.4)

C-2 `src/ap_forecast/explain.py` *(changed, TSK-04)*

Only the signature line and one conditional statement change; `load_config`, `CONFIG` and the returned dict stay as they are.

```
def build_explanation_request(vendor: str, predicted_days: int, config=None, risk=None) -> dict:
    cfg = config or load_config()
    prompt = f"Explain in one sentence why vendor {vendor} is predicted to be paid {predicted_days} days late."
    if risk is not None:
        prompt += f" Risk level: {risk}."
    return {...}  # unchanged
```

Rule	Why	Source
`risk` is the last parameter, default `None`	Existing positional and keyword calls stay valid (AC-04.3)	feat.trd (IF-4.1); feat.stories (AC-04.3); file:ap_forecast/CLAUDE.md ("Keep existing public signatures")
"Given" means `risk is not None`; any non-`None` value is appended without validation	Interim rule pending TQ-4 / CQ-4	feat.trd (IF-4.2, IF-4.5); journey:feat.clarifications (CL-DS5)
Provider, model and `max_tokens` unchanged	AC-04.4	feat.stories (AC-04.4); feat.trd (IF-4.4)

C-3 `src/ap_forecast/__init__.py` *(changed, TSK-03)*

```
from .predict import predict_delay_days
from .explain import build_explanation_request
from .risk import vendor_risk

__version__ = "<minor step from 0.9.0, Developer to confirm>"
__all__ = ["predict_delay_days", "build_explanation_request", "vendor_risk"]
```

Existing exports stay. The version step is owned by TSK-03 and shared with TSK-04, so it is made once (feat.tasks section 2, TSK-03 "Done when", assumption TA-3 in section 6; feat.design DD-11). The exact number is not given in any input (open question DSQ-1).

C-4 `tests/test_risk.py` *(new, TSK-03)*

Test	Asserts	Source
`test_vendor_risk_levels`	Histories predicting 31 → `'high'`, 20 → `'medium'`, 5 → `'low'`	feat.design (B2.5); feat.stories (AC-03.1, AC-03.3)
`test_vendor_risk_boundaries`	Single-value histories `[30]` → `'medium'`, `[31]` → `'high'`, `[10]` → `'low'`, `[11]` → `'medium'`	feat.tasks (TSK-03); journey:feat.clarifications (CL-DS1); record:memory (L-007)

Test	Asserts	Source
`test_vendor_risk_empty_and_early`	`[]` → `'low'`; only negative delays → `'low'`	feat.design (B2.5); feat.stories (AC-03.5, AC-03.6)
`test_vendor_risk_no_llm`	Patch `ap_forecast.explain.load_config` to fail; `vendor_risk` still returns	feat.tasks (TSK-03); feat.stories (AC-03.7)
`test_vendor_risk_exported`	`"vendor_risk" in ap_forecast.__all__`	feat.design (B2.5); feat.stories (AC-03.8)

Agent judgement on the C-1 sketch: the boundary histories `[31]`, `[30]` and `[10]` reach each of its return statements, so the new module should add no uncovered line. CI measures coverage over the whole package, so a new module without tests would lower it (file:ap_forecast/ci.py; journey:feat.clarifications CL-X6).

C-5 `tests/test_forecast.py` *(changed, TSK-04)*

Test	Asserts	Source
`test_request_with_risk`	Prompt ends with `' Risk level: high.'`	feat.tasks (TSK-04); feat.stories (AC-04.1)
`test_request_without_risk_unchanged`	Result equals today's, with and without explicit `risk=None`	feat.tasks (TSK-04); feat.stories (AC-04.2, AC-04.5)
Existing three tests	Pass unchanged	feat.design (B2.5); tool:repo (`tests/test_forecast.py`)

4. Components and layers that do not change

Component / layer	Status	Source
`src/ap_forecast/predict.py`	Unchanged	feat.tasks (TSK-03 "Must not change"); feat.design (B2.1)
`src/ap_forecast/models.json` (LLM provider configuration)	Unchanged; provider `acme-llm`, model `acme-text-2`	feat.tasks (TSK-03, TSK-04); tool:repo (`models.json`)
`ci.py`, `README.md`, `CLAUDE.md`	Unchanged; not in either task's file budget	feat.tasks (TSK-03, TSK-04 "Files in budget"); file:ap_forecast/ci.py; file:ap_forecast/README.md
Database / storage	None in this repository and none required	journey:task.db_changes (sections 1, 3); feat.design (A1, DD-1)
Secrets	None required or granted for the `ap_forecast` repository; the code builds the LLM request but does not send it. The permission table answers per repository, not per environment; confirming every environment is the open question SSQ-2	journey:task.secrets_scope (sections 1, 2, 3)
Other repositories (`ap_pipeline`, `ap_aging`, `ap_portal`)	Not touched by this lane; no import of another lane's package	journey:feat.clarifications (CL-X2); feat.design (Part C "Lane dependencies")
Transport, service, new library	None; standard library only	feat.design (DD-1, DD-3)

5. Order and parallel work inside the lane

TSK-03 and TSK-04 touch different files (C-1, C-3, C-4 versus C-2, C-5), and neither depends on the other in code, so they can be built in either order on the existing `feature/story` branch (feat.tasks section 1; task.repo_access RAQ-1). Commit messages start with the story ID, `STORY-03` or `STORY-04`, and the task ID may follow (feat.tasks section 2). TSK-04's release, not its coding, waits on the order against PROV-ACME-2026-11 (PQ-5) and the Architect's answer to TQ-8 (feat.tasks TSK-04; journey:feat.clarifications CL-DS7, CL-DS8).

6. Constraints carried into DS-05 and DS-06

Constraint	Source	
Python 3.9 syntax only (no `match`, no `X \ Y unions); typing.Iterable`		feat.tasks (section 2); feat.design (NF-9)
No `print`, `eval`, `exec` or bare `except`; lines at most 140 characters	file:ap_forecast/ci.py; feat.tasks (section 2)	
Coverage of at least 90% over `src/ap_forecast`	file:ap_forecast/ci.py; feat.tasks (section 2)	
Keep existing public signatures; add new functions for new behaviour	file:ap_forecast/CLAUDE.md; feat.design (header "Architecture rule")	
Synthetic test data only; no secrets or personal data	feat.trd (S-6); journey:feat.clarifications (CL-X9)	

7. Note on an input difference

feat.stories TK-03.2 proposes adding the risk tests to `tests/test_forecast.py`, while feat.tasks (section 2; assumption TA-4 in section 6) puts them in a new `tests/test_risk.py`. feat.design B2.5 and feat.clarifications CL-X4 allow either location. This plan follows feat.tasks, the later output, which the journey board shows as approved by the Developer at DES-04. feat.tasks still lists TKQ-2 (confirm the test-file split) as a non-blocking question for the Developer, so the location can move back to `tests/test_forecast.py` without any change to C-1 to C-3 (feat.stories TK-03.2; feat.tasks sections 2, 6, 7; feat.design B2.5; journey:feat.clarifications CL-X4; record:memory).

8. Open questions (none blocking)

ID	Question	Interim answer	Owner (proposed)
DSQ-1	Which version number replaces `0.9.0`? The design asks for a minor step for the lane to confirm, and no input gives the number	DS-05 proposes the minor step; the Developer confirms it at review	Developer

Carried, unchanged: TQ-4 and CQ-4 (meaning of "given" for `risk`), TQ-5 (early payments counted as 0), TQ-8 (risk label sent to `acme-llm`), PQ-1 (empty history), PQ-5 (order against PROV-ACME-2026-11), RAQ-1 to RAQ-3 (task.repo_access), SSQ-1 and SSQ-2 (journey:task.secrets_scope). None blocks coding (journey:feat.clarifications section 10).

9. Validation check

Check	Result	Source
Existing code read and understood	Pass (section 2)	tool:repo; file:ap_forecast/ci.py; mcp:codegraph
Every lane task mapped to components	Pass: TSK-03 → C-1, C-3, C-4; TSK-04 → C-2, C-5	feat.tasks (section 3)
Changes stay inside each task's file budget	Pass	feat.tasks (TSK-03, TSK-04)
Minimum change: no existing signature, return shape or file outside the budget changed	Pass	feat.design (B2.1, B2.2); file:ap_forecast/CLAUDE.md
Secrets	None required for the repository; the per-environment confirmation stays open as SSQ-2 (non-blocking)	journey:task.secrets_scope
Database changes	None required	journey:task.db_changes
Decision log and lessons (L-007 boundary tests)	Consistent	record:memory
Blocking questions	None	—